

MPONLINE IDEA & INNOVATION HACKATHON 2026



TITLE PAGE · INTRODUCTION

Challenge – 5: Healthcare

Problem Statement – Emergency patients lose the golden hour – families cannot see which hospital can treat them right now, and citizens, hospitals and 108 are not connected.

Solution Title – MEDREACH – AI-powered emergency coordination for Madhya Pradesh, integrated with the 108 / 102 national ambulance services

Theme – Healthcare · Emergency response

Category – Software (web app + integrations)

Team Name – [TEAM MEDREACH]

Live Prototype – medreach-2.onrender.com

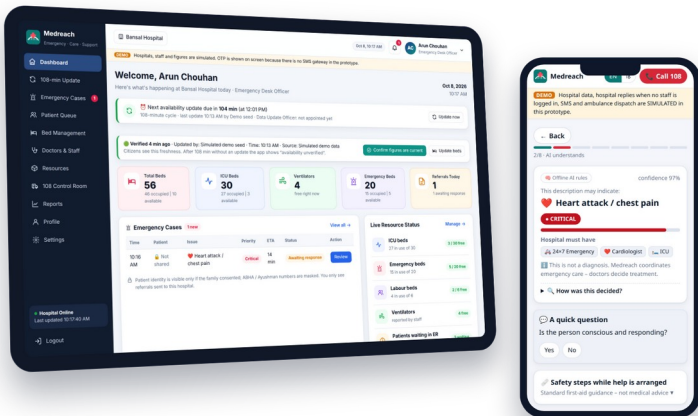


The right hospital, right now – confirmed before you leave home.



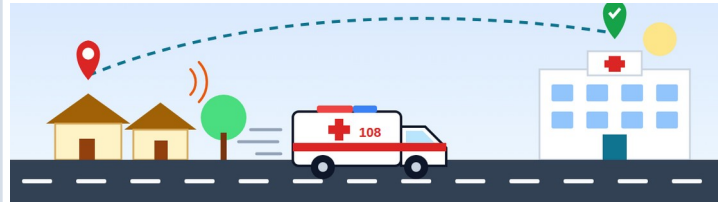
THE EMERGENCY

- ✦ 9:40 PM, a village near Berasia: Ramesh's father (58) has chest pain and sweating.
- ✦ Only 2G works; the town hospital is 25 km away.
- ✦ Nobody knows which hospital has a cardiologist and a free ICU bed tonight.
- ✦ Calling 108 means repeating the address on a weak line.



From Guesswork to Guaranteed Care

Right hospital | Confirmed bed | 108 linked



Today

With Medreach

1 FINDING A HOSPITAL

Drive to the nearest one

Only hospitals that can treat it

2 CONFIRMING A BED

Refused at the gate

Hospital accepts first

3 CALLING 108

Repeat address, call drops

Signed incident sent

4 INFORMING THE ER

Arrives unannounced

AI pre-arrival note

5 TRACKING

No idea where it is

Live map + ETA

The Result **Guesswork** → **Confirmed bed**

WHAT MEDREACH DOES

- ✦ Understands Hindi or English voice in seconds.
- ✦ Shows only hospitals with the right specialist, bed and blood – live.
- ✦ Hospital accepts and reserves a bay before the family leaves.
- ✦ Hands the case to 108 / 102 and tracks the ambulance.

Why Medreach?



Right hospital



Accept first



108 linked



Works on 2G



Secure data

Every emergency, a confirmed bed

THE PROBLEM

1,68,491

road-accident deaths in India in 2022

Source: MoRTH, Road Accidents in India 2022

Heart attacks, strokes, burns and childbirth emergencies face the same delays.

ROOT CAUSES

- Nearest hospital $\neq$ capable hospital
- Bed and doctor data is old – nobody owns it
- No acceptance before the patient arrives
- Villages have only 2G or no data

WHO IS AFFECTED

- Patients and families in panic
- Rural citizens and ASHA workers
- ER teams facing surprise arrivals
- 108 call-takers re-asking details

1 CITIZEN SPEAKS



CITIZEN APP

- Voice in Hindi / English
- Aadhaar + OTP sign-in
- Live beds, doctors, blood

2 MEDREACH MATCHES



HOSPITAL DASHBOARD

- Accept & reserve a bay
- AI pre-arrival (SBAR) note
- 108-minute data updates

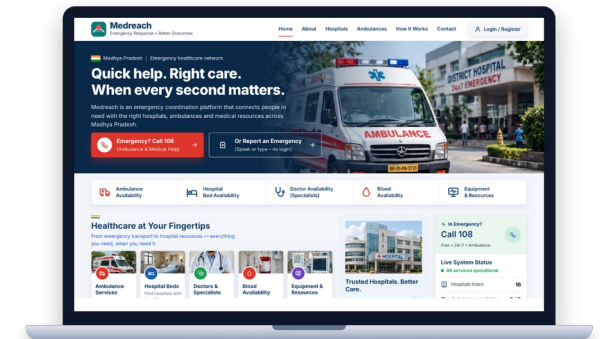
3 HOSPITAL ACCEPTS



108 CONTROL ROOM

- Signed incident, no re-asking
- Nearest suitable unit
- Live status to the family

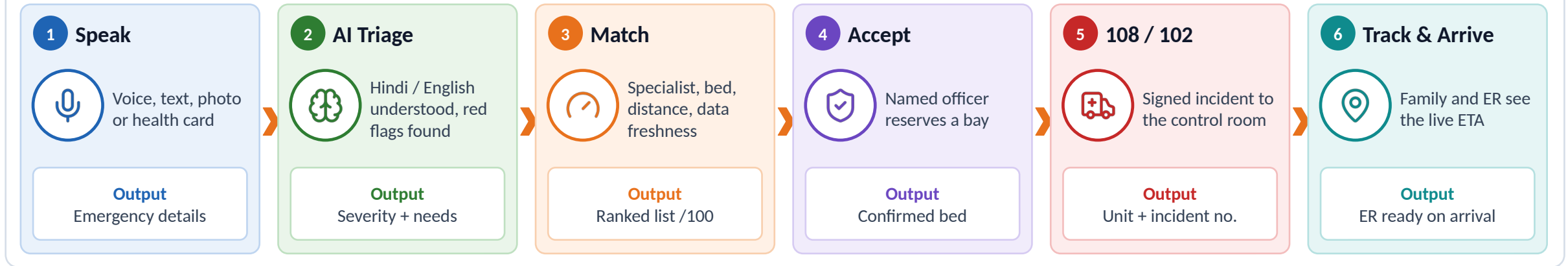
4 108 DISPATCHES



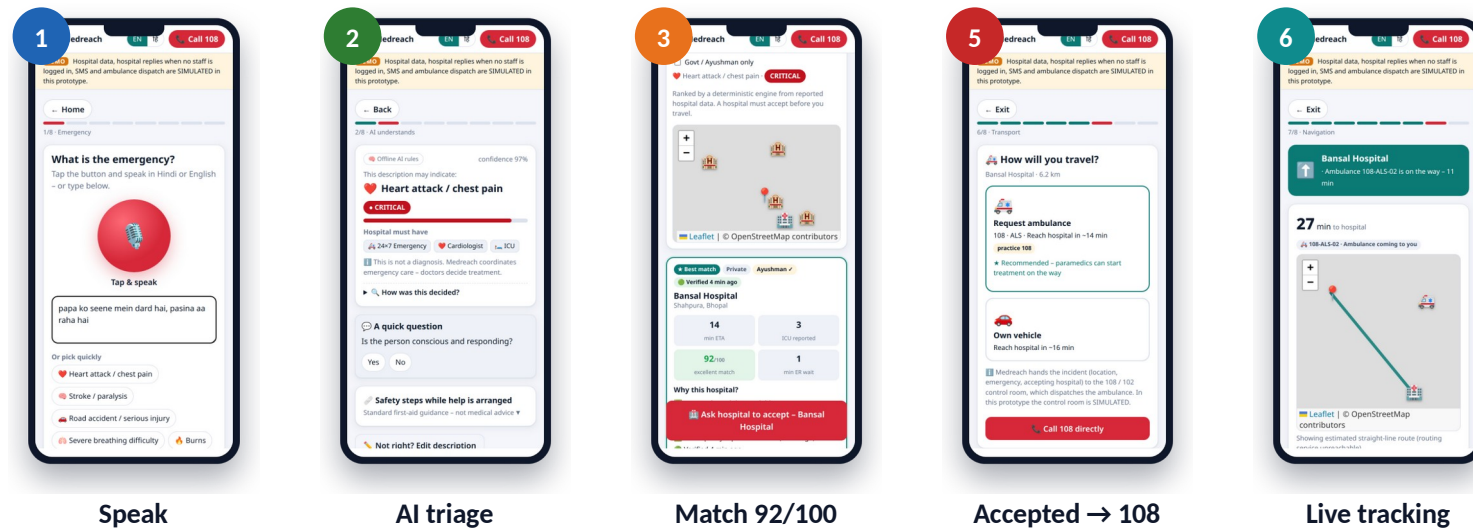
Live home page

MEDREACH = ONE LIVE LINK BETWEEN CITIZEN, HOSPITAL AND 108

THE COMPLETE FLOW: From First Words to a Confirmed Bed



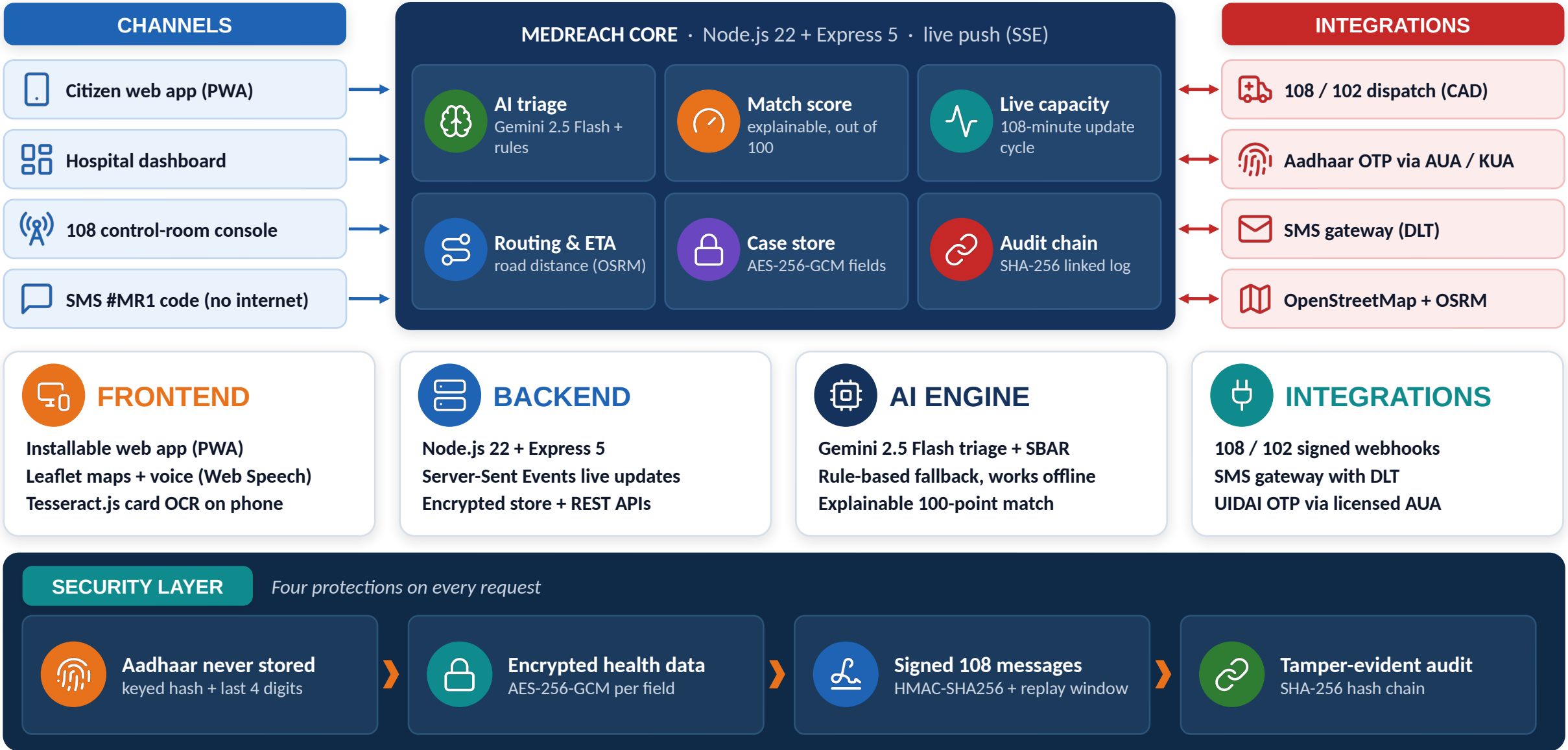
REAL SCREENS FROM THE LIVE PROTOTYPE



RULES WE NEVER BREAK

-  Hospital accepts before anyone travels
-  Only hospitals that can treat the case
-  60 s without an answer → next hospital
-  Never diagnoses – says “may indicate”
-  Call 108 is visible on every screen

TECHNICAL APPROACH



BUILT & WORKING

- Live website – no app store needed
- Citizen app, hospital dashboard, 108 console
- 94 automated tests · Hindi + English
- Open-source stack – no licence cost

KEY RISKS

- Hospitals may not keep data current
- Weak or no network in villages
- Each state's 108 runs its own system
- Health and Aadhaar data privacy

MITIGATION

- 108-min cycle, named officer, “unverified” flag
- SMS gateway, offline app, 108 / 112 call
- Open signed contract + practice control room
- AUA/KUA OTP, AES-256-GCM, consent

1 PROTOTYPE LIVE

2 DISTRICT PILOT

3 MP ROLLOUT

4 NATIONAL 108 / 102

PROTOTYPE TODAY

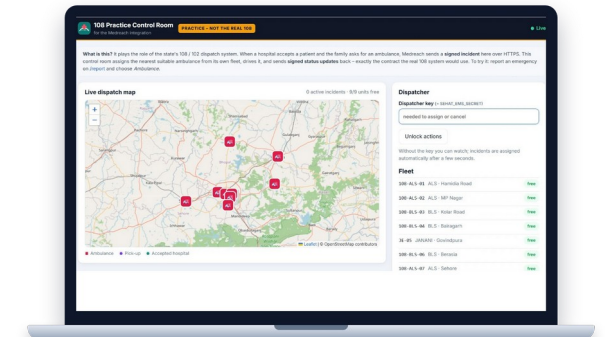
- 16 hospitals · 4 districts (sample data)
- 9-ambulance practice 108 fleet
- Aadhaar OTP (simulated UIDAI)
- Live: medreach-2.onrender.com

NEXT PILOT

- Bhopal division with CMHO + 108 operator
- Real Data Update Officers
- Licensed AUA/KUA + DLT SMS
- Measure time to confirmed bed

SCALE

- All districts of Madhya Pradesh
- Any state's 108 / 102 – same contract
- Capacity analytics for health dept.
- ABHA health-record link



Live 108 control room (deployed site)

FEASIBLE TODAY: SOFTWARE ONLY, RUNS ON PHONES AND HOSPITAL PCs ALREADY IN USE

TODAY vs MEDREACH

Capability	Calling around	Map apps	Medreach
Specialist & ICU free right now	maybe	✗	✓
Hospital confirms before travel	sometimes	✗	✓
Speak Hindi, AI understands	✓	✗	✓
108 gets location + hospital	✗	✗	✓
Works on 2G / SMS / offline	✓	✗	✓
Data owner, 108-min freshness	✗	✗	✓
Verified users, tamper-proof log	✗	✗	✓

KEY INNOVATIONS

- ✦ Accept before travel: no family leaves without a hospital's "yes".
- ✦ Explainable match score out of 100 – every part shown.
- ✦ 108-minute update cycle with a named Data Update Officer.
- ✦ Signed 108 hand-off any state's control room can adopt.
- ✦ A whole emergency fits in one SMS (#MR1 code) on 2G.

VOICE AI

MATCH SCORE

ACCEPT FIRST

108 LINKED

★ Best match Private Ayushman ✓
Verified 4 min ago

Bansal Hospital
Shahpura, Bhopal

14 min ETA	3 ICU reported
92/100	1

Match score 92/100

explainable ranking

❤️ Heart attack / chest pain Critical Awaiting r

Family did not consent to share patient details

ETA 14 min ICU bed transport: deciding note: te

S: Unknown patient – CRITICAL Heart attack /
B: No history available.
A: CRITICAL – Heart attack / chest pain (adult)
R: Prepare 24x7 Emergency, Cardiologist, ICU.

AI pre-arrival note

SBAR reaches the ER first

liability update
correct Data Update Officer confirms everything citizens, 108 and the matching engine rely on.

liability update due in 104 min (at 12:01 PM)
last update 10:10 AM by Demo user - Data Update Officer: not appointed yet.

Patients waiting in ER	ER doctors on shift
3	3

ad-only for your role

Emergency free at 20	Labour free at 10
5	2

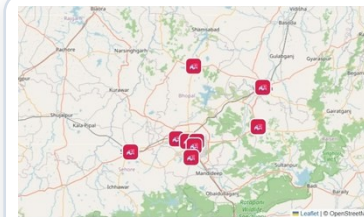
A Data Update Officer not appointed yet.
The Nodal Officer appoint keep these figures current.

Reminders

- 10 min before the update
- When it is due (please on)
- 15 min late: overbook, etc

108-minute update

named Data Update Officer



108 control room

live dispatch map

Try the SMS channel demo
This is the exact SMS the app writes when there is no internet. In the demo it goes straight to our server instead of through an SMS gateway – the server then does exactly what it does for a real SMS.

Emergency in a village near Beraisa

Heart attack – chest pain

MEDREACH SOS: Heart attack / chest pain (CRITICAL) at 23.60123,77.40456
#MR1 cardiac C 23.60123 77.40456 - en

MEDREACH SS-FBF658: Heart attack / chest pain (CRITICAL) received. Asking CHC Beraisa to accept now – you will get an SMS. If the patient gets worse, CALL 108.

Open live tracking →

SMS #MR1 code

works on 2G

Sign in or register with Aadhaar + OTP. One verified account per person, so fake accounts can't be created.

OTP sent to the mobile linked with Aadhaar XXXX XXXX 0013: *****1001. Valid for 5 minutes.

Demo only (simulated UIDAI, no SMS is sent): your OTP is 156148

6-digit OTP *

Aadhaar + OTP

one person, one account

IMPACT AND BENEFITS



CITIZENS

Right hospital the first time
Speak - no forms in panic
Track the ambulance live



HOSPITALS

Patients arrive announced
Bay ready, SBAR note first
Fewer unsuitable walk-ins



108 & GOVT.

Complete incident, no re-asking
Live capacity map of MP
A data owner for every hospital



RURAL MP

Works on 2G, SMS and offline
102 Janani Express for mothers
ASHA workers can report

◆ KEY BENEFITS OF CONNECTED EMERGENCY CARE ◆



FASTER DECISIONS

Guesswork
→ **Confirmed**

- Hospital must answer in 60 s
- Ranked list in seconds



RIGHT HOSPITAL



- Only hospitals that can treat it
- Fewer second trips



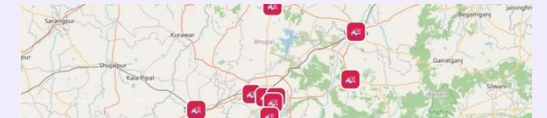
LOW COST

No new hardware ✗

- One web app for every phone
- Open-source stack, AI optional
- Runs on one small cloud server



SMARTER PLANNING



- Live capacity across districts
- Real demand data for planning

EXPECTED OUTCOME

Every emergency in Madhya Pradesh reaches a hospital that has already said **yes** – with 108 on the way and the ER ready – **confirmed in minutes, not after a wasted trip.**



It works today

Live site: speak → hospital accepts → 108 ambulance → tracking, end to end.



Ready for real 108

Signed incident contract, proven with our practice control room.



Built for rural MP

Hindi voice, 2G / SMS / offline, 102 Janani Express for mothers.



Secure & private

Aadhaar OTP, AES-256-GCM, tamper-evident audit, no Aadhaar stored.



Low cost, scalable

Open-source stack, one web app for every phone, AI optional.



Responsible AI

Never diagnoses; hospitals decide; Call 108 is always one tap away.

SCAN TO TRY IT LIVE



medreach-2.onrender.com

Code: github.com/prakharpurwal518-collab/Medreach

✓ WORKING PROTOTYPE

✓ 108 INTEGRATED

✓ RURAL-READY

✓ SECURE BY DESIGN

THANK YOU · धन्यवाद

Medreach – the right hospital, right now, confirmed before you leave home.